


# Conceptual Design Report 1

## Presented to Up Goers Ltd. for the Mars Sample Return Program (MSRP)

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# Executive summary

On September 6th, 2021, NASA's Perseverance rover successfully collected and stored a Martian rock sample in a sealed titanium tube, marking an important step in advancing the Mars Sample Return (MSR) program. Following the event, more samples were then collected and stored onboard Perseverance in the following weeks. To provide redundancy in case of the rover's failure or misplacement, ten samples were deposited at the Three Forks Depot (TFD) site.

UpGoers Ltd., a Canadian space contractor, is currently planning to launch a sample retrieval mission which will send a retrieval lander to Mars equipped with a rover that will collect the samples left at Three Forks Depot. The rover will then store the samples in a Mars Ascent Vehicle which will place the samples in orbit around Mars for recovery and transport back to earth for scientific analysis. Returning Martian samples to Earth allows the use of cutting-edge equipment for sample analysis, furthering our understanding of life on Mars. The MSR project involves daunting engineering challenges, requiring innovative solutions every step. The sample retrieval rover must operate in the harsh Martian climate (temperatures from 20°C to -150°C, up to planet-wide dust storms, etc.), navigate safely to the samples by avoiding obstacles in its way, and physically retrieve the samples from perseverance, and the TFD, and load them onto the Mars Ascent Vehicle. Ptarmigan is developing a prototype rover to demonstrate their ability to provide a novel and practically functional wheeled rover to UpGoers Ltd. for the sample retrieval mission. UpGoers Ltd. requires the prototype to meet the following system level functionalities: a form of navigation, some form of sensor integration, and a human robot interface allowing control and monitoring which is ideally user friendly and capable of seamless interaction between the rover and the mission operators on Earth. UpGoers Ltd. encourages the prototype to showcase new and innovative ways to aid in scientific analysis of Mars in addition to the primary requirements.

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# Introduction

The Mars sample retrieval mission is a joint effort between NASA and ESA that has been in progress for many years. UpGoersLTD plans to complete a crucial phase in this mission by launching the samples from mars into orbit for transport back to earth. Ptarmigan aims to provide a rover capable of sample recovery from the samples deposited at three forks depot to maximize scientific gain from this mission.

Within the Mars sample return architecture, surface retrieval rovers play a critical role by locating, acquiring, and transferring cached sample tubes while maintaining strict planetary protection requirements. Previous Mars rover missions have demonstrated both the feasibility and the risks of long-duration autonomous surface operations, including mobility hazards such as the embedding of the Spirit rover and the increasing reliance on onboard autonomy exemplified by the Perseverance rover.

This report presents the design and operational details of the mars sample retrieval rover, including the current progress of hardware and software development of the prototype, current tests and simulations, and the intended implementation.

Ptarmigan has identified and solved key challenges for this mission including robust mobility in uncertain terrain, reliable autonomous manipulation under communication delay, and contamination-controlled sample handling. Ptarmigan's design of the rover not only ensures mission success, but also strictly follows regulations and standards laid out by COSPAR Category V planetary protection requirements.

# Technical Engineering Progress

## Hardware

The rover prototype assembly has reached its final state for the extent of this part of the design task. It is partially assembled, with the computer, LIDAR and IR systems being installed and wired, while the mounting system as well as robotic arm have left to be prototyped and deployed onto the rover.

Ptarmigan will employ the CSA SPRP (Small Planetary Rover Platform) as a cost effective alternative to developing a native platform. This cooperation with the CSA will allow Ptarmigan to ensure mission success on time, and within budget, while still building on proven Canadian Space hardware. This was deemed especially important on a low budget-tight timeline mission such as the MSR mission. Effectively, this allows Ptarmigan to focus development resources on the sample retrieving device cache system.

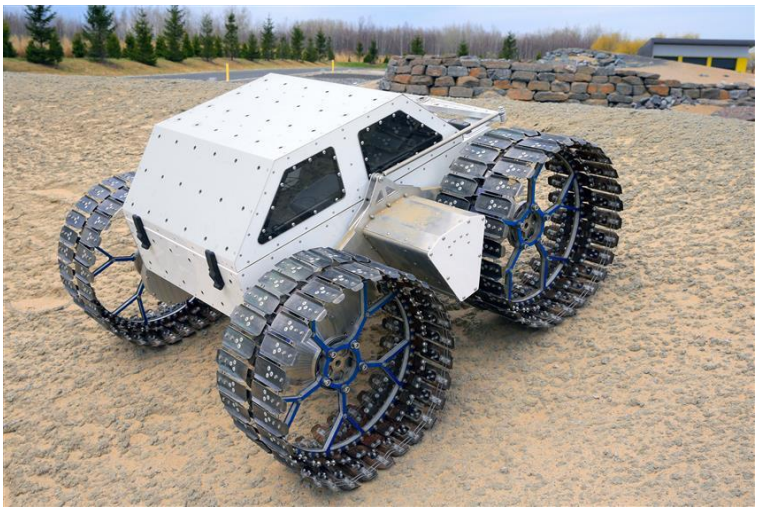


*Figure 1: SPRP Rover Courtesy: Canadian Space Agency [7]*

# Software

The hardware running software systems include a Raspberry Pi 4 (RPi) computer as well as an Arduino Uno Rev2 WIFI. Given the task requirements and the clear need for some level of autonomy, Ptarmigan Aerospace has decided to integrate ROS2 into the rover design to handle mapping the environment, locating itself within that environment, and autonomously navigating to its next waypoint while avoiding obstacles on its path. The Nav2 library has been chosen for SLAM as it is well established and extensive documentation and troubleshooting guides are available. The Gazebo and Rviz applications have been selected to act together to simulate the rover's movements and actions, allowing Ptarmigan to identify and address bugs proactively and reduce the risk of failures once the rover is tested in a real-world scenario. Once on the Martian surface, the software systems are intended to identify and mark all sample tubes as waypoints. Waypoints will be set automatically by the onboard computer and will be updated each time a sample is picked up with updated locations based on rover scans of the surface.

Given the fact that the RPi has greater computation power than the Uno, it was chosen to handle the implementation of ROS2 and running its associated libraries. Ptarmigan's intention is for the RPi to compute simultaneous location and mapping calculations using odometry readings and output wheel speed inputs over a serial connection to the Arduino Uno, which then will convert the speed data into signals to the motors.

Ptarmigan setup the onboard Raspberry Pi operating system through a wired connection to a host system in order to implement Secure Shell (SSH) communication. In parallel, Ptarmigan installed and setup a Virtual Machine (VM) running Ubuntu 22.04 (the same version of Linux on the RPi) on the host computer to run the simulation software. ROS2, Rviz, and Gazebo systems were the next programs to be installed on both the client (RPi) and host (Ubuntu VM) machines. Running and testing Gazebo proved to be the most difficult part yet, as its implementation on a VM was riddled with bugs and errors, mostly with respect to the VM's OpenGL version being too outdated (v1.9) for the Modern Gazebo version (requiring v3.3+), and there being conflicting evidence on if a newer, compatible version of OpenGL was even possible to install on the host VM configuration. Multiple hours were spent debugging this part of the system, but finally OpenGL v4.0 was installed by force on the host system and that bug was cleared. Other errors such as the "Playground" module not being authenticated properly were gone around by launching gazebo without the module, which led to a negligible performance change and thus was ignored in order to save time and move on to more pressing tasks.

## Testing and Simulation

Simulation has had an extensive role in the development and validation of the rover's software. A Gazebo-based simulation environment was successfully established, running on the host VM, providing a relatively stable platform for visualizing and testing rover behaviour. Basic simulating functionality was confirmed by launching standard and empty simulation worlds and verifying consistent performance, ensuring the environment is suitable for continued development and testing.

Following this, efforts focused on integrating the rover's digital model into the simulation software. The rover's Unified Robot Description Format (URDF) model was built and incorporated into ROS2, with extra debugging required to correct package structure and installation paths so that the model can be reliably discovered and launched from the ROS package share directory. This involved resolving many integration challenges such as build-time installation errors, launch file configuration issues, and ensuring the simulator references the installed model rather than source-only files. Successfully addressing these issues will empower us to spawn the rover within many simulation worlds.

The simulation groundwork that is currently being laid will significantly reduce technical risk by identifying integration issues early and ensures that future autonomy features can be developed and tested in a controlled, repeatable manner.

The next phase of simulation work will focus on implementing autonomous navigation using the ROS2 Navigation Stack. This will enable the rover to localize itself, plan paths, and navigate on the Martian surface while avoiding obstacles, all in real time. The objective of this work is to allow the rover to independently navigate its environment to reach sampling locations. Upcoming simulation tasks include integrating navigation sensors (such as odometry and LIDAR data), generating and using testing environment maps, tuning navigation parameters for offworld terrain, and validating endtoend autonomous sample approach behaviors prior to physical testing.

# Implementation and Deployment Outline

## Operational Overview

The rover will employ multiple systems to effect sample retrieval. An overview of the main systems is provided below, namely, the rover chassis, drivetrain, power collection, retrieval arm, retrieval claw, and storage bank.

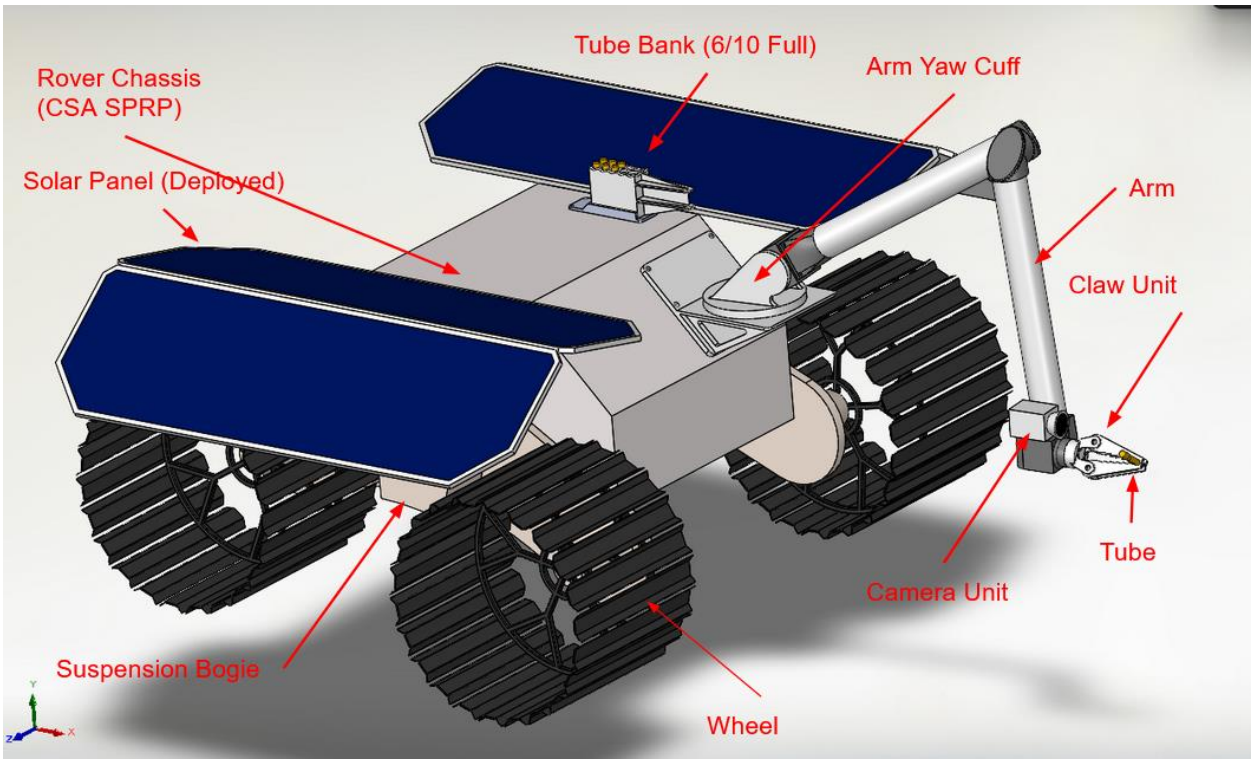



*Figure 2: Rover design Overview, with highlighted components. CAD by Ptarmigan engineer.*

After being delivered to Mars via the provided landing system, the rover will engage in a set of steps to begin mission operation. Once delivered near the Three Forks Depot, the rover will receive directional instruction from the lander and associated craft. After the rover leaves the lander, it will proceed through Startup, Operation, Conclusion, and Shutdown as outlined below.

## Startup Procedure

The rover is designed to stay in a stowed (See below) state in the lander during its flight to Mars. Deorbit, and landing. After landing on Mars, it will power on within the lander, initialize sensors and data, and receive map orientation information from the landing craft. During this period, it will communicate, through the landers long range communications to receive necessary data and

instruction from mission control. After initialization, the rover will proceed to untether from the craft, unfold solar panels, and activate its arm. Its mobility system and autonomous control will be checked and calibrated, if necessary, by performing short distance autonomous movements. Once the rover is verified to be in working order, it will wait for the initiation command from control002E

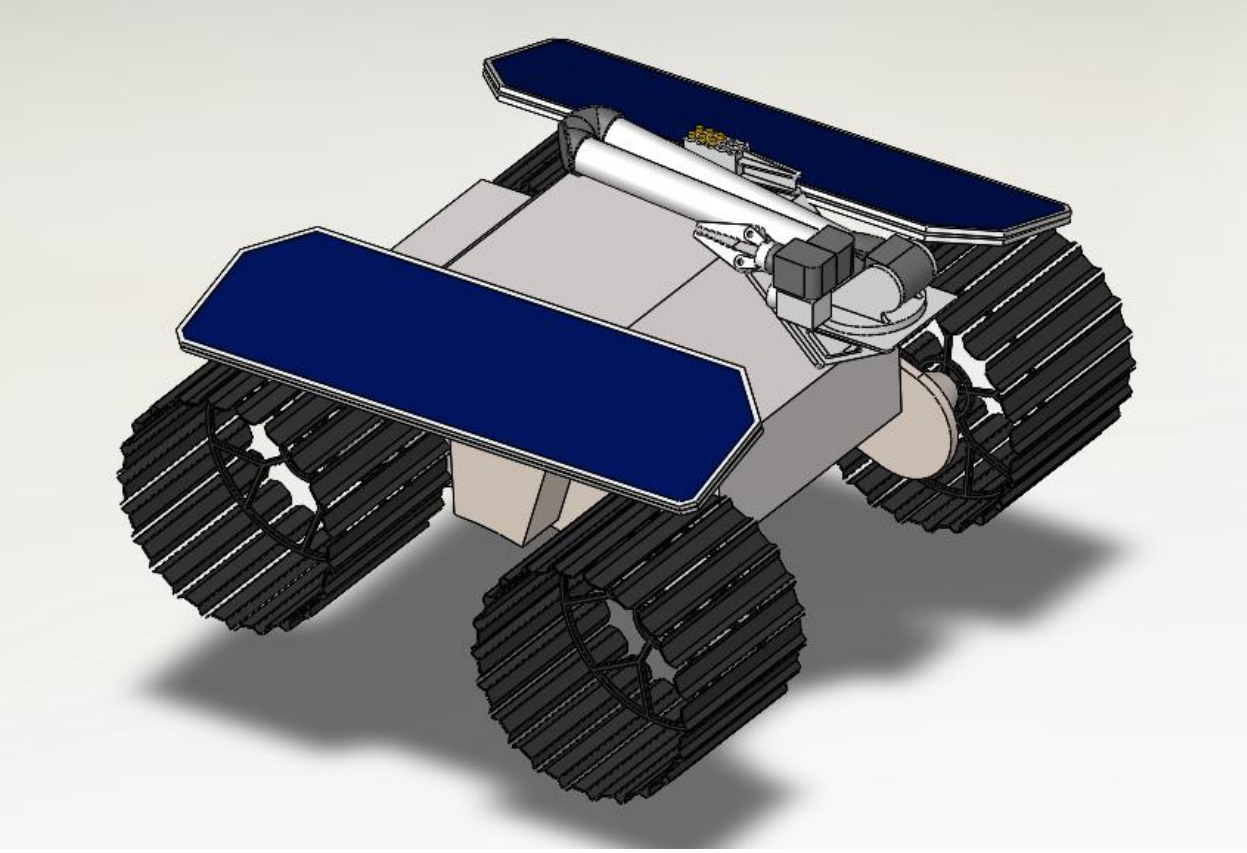


*Figure 3: Rover in Stowed Configuration*

## Surface Operations

The rover must travel from its current location to the first tube; the location will be determined by mission control on earth and transmitted to the rover which will autonomously navigate towards it while avoiding hazardous terrain such as sand, slopes and rocks. Due to communication delay, the rover will be provided waypoints to traverse to rather than immediate controls. The rover with employ an autonomous pathfinding algorithm to achieve this end.

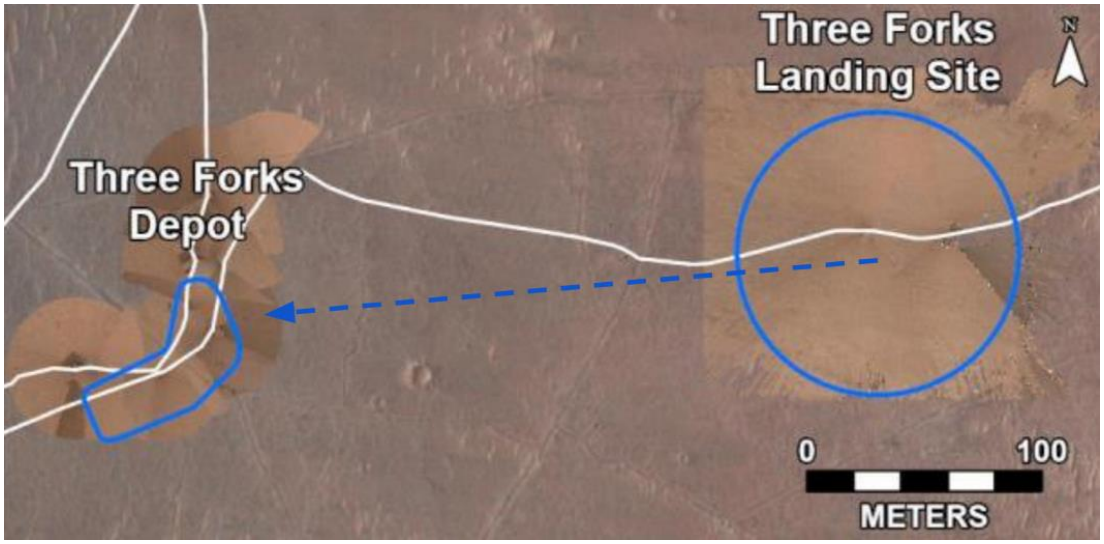



*Figure 4: Three Forks Landing Site and Sample Depot, with Travel Direction Identified. Courtesy: NASA [6]*

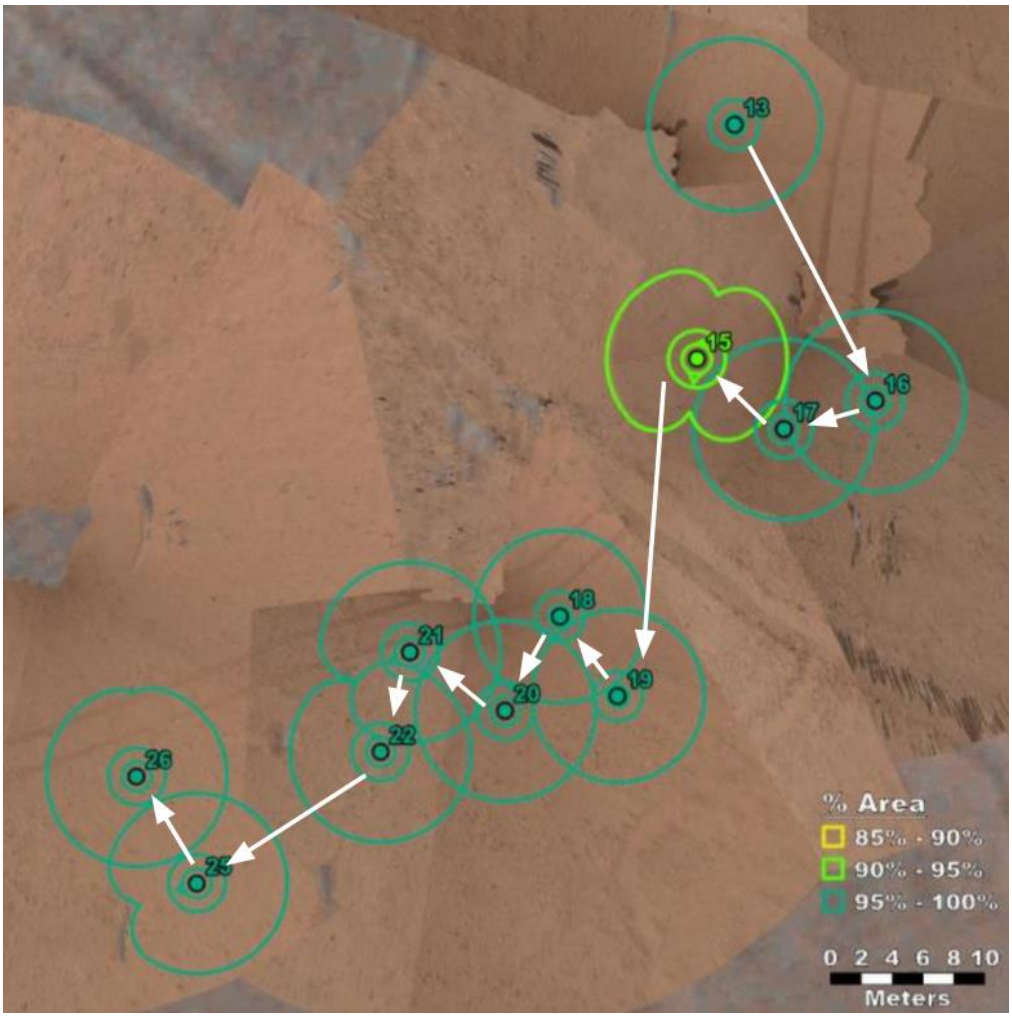



*Figure 5: Sample position data retrieved following the Mars 2020 mission. Courtesy: NASA [6]*

When close to the tube, the rover must locate its exact position and orientation using on board camera and align its approach to match the tube's orientation and position with data from on board IMU sensors. If the rover fails to locate the tube, it will change its viewing angle and try again until the tube is located. When close to the tube, the rover will execute a slow, closed-loop final approach that maintains a keep-out zone to prevent driving over the tube and compromising sample integrity and align it's retrieval mechanism with the tube. The retrieval mechanism will grab the tube and guide it into the storage bank without damaging any seals. Sensors inside the storage chamber confirm sample acquisition. On-board inventory is updated before the rover proceeds to the next retrieval cycle.

## Mission Completion, Handover and Shutdown

Upon collection of sample tubes, the rover will proceed back to the landing craft and complete the "handover" maneuver. The rover will use its multi-use arm to grip and remove the sample bank from its chassis stowage slot, and transfer it into the escape craft.

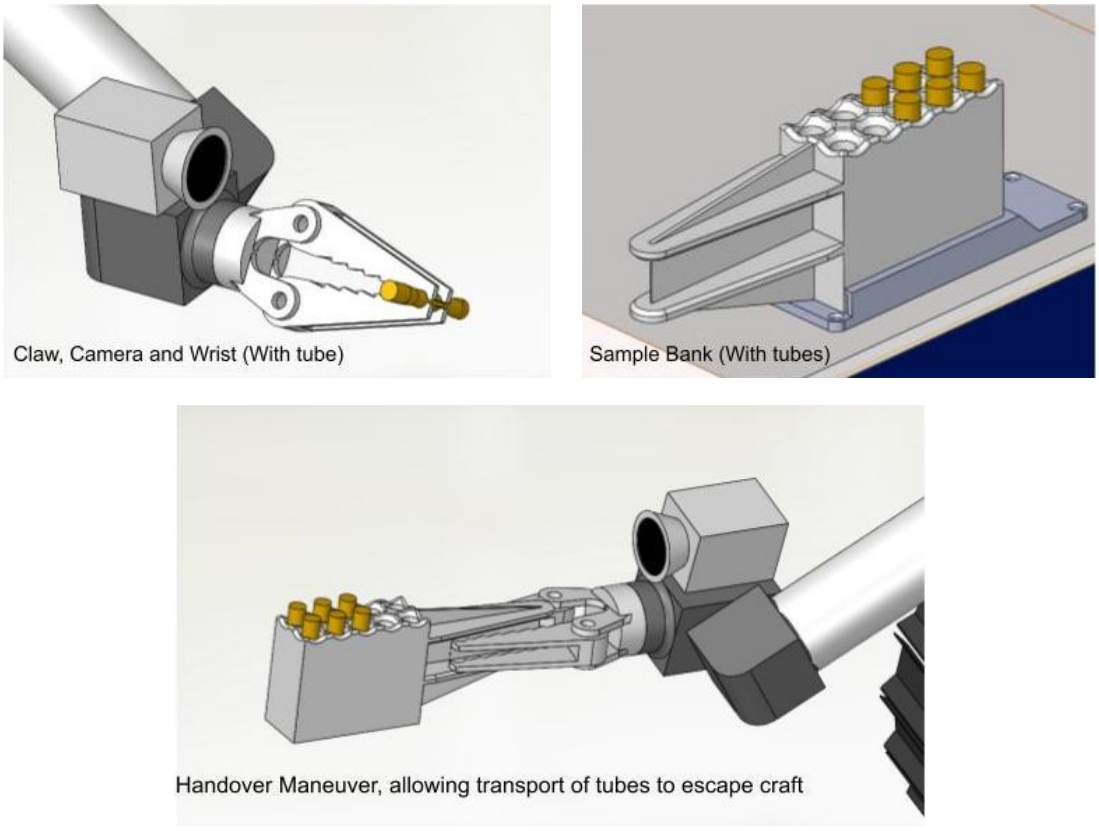



*Figure 6: Claw and Bank assemblies*

After handover is confirmed successful by communication with the lander, the rover will proceed to evacuate the immediate launch area, and sleep. The rover will remain in contact with the lander and Earth for potential re-issue of a new mission objective in the future.

## Fault Management and Contingencies

Ptarmigan has incorporated contingency handling and fault awareness informed by lessons learned from prior Mars surface missions. The rover is designed to detect anomalies, diagnose root causes, and autonomously execute recovery actions or transition to a safe state as appropriate.

A failed sample grasp is detected through force and position feedback, triggering a controlled back-away and retry sequence. If a sample tube cannot be immediately localized, the rover executes a visual search before requesting ground assistance. The rover uses a Mobility Fault Protection protocol that automatically stops itself if it's outside its preset safety limits, preventing damage to motors or driving on risky steep tilts. Moreover, it features continuous slip monitoring, odometry mismatch detection and an autonomous back out protocol when slip limits are exceeded to prevent the rover from being embedded in sand, a protocol derived from the event that caused the spirit mars rover to be permanently stuck in soft sand [4].

In the event of extended communications delay or loss, the rover transitions to a safe standby state where it consumes low power and monitors system status. Potential, known causes of this include dust storms, solar storms, or equipment faults.

## Operational Assumptions and Constraints

Ptarmigan has designed the rover to operate in daytime conditions with solar/battery power, to avoid the possibility of failing to detect hazardous obstacles. The rover will be constrained by a maximum slope height in the terrain, terrain roughness, maximum rock height, minimum temperature, and expected dust exposure. To facilitate sample retrieval, Three Forks Depot was chosen due to its terrains low slope, low roughness, and low max rock height, therefore the rover is designed for moderate terrain consistent with the environment at three forks depot. Moreover, exactly like Perseverance, the rover is designed for typical Martian dust exposure with mitigation for particulate ingress, not full burial. [5]

# Non-Technical Progress

## Regulatory Standards

Ptarmigan is considering several regulations governed by the Outer Space Treaty, COSPAR Planetary Protection Policy, NASA/ESA mission requirements, and Sample Receiving Facility constraints. These regulations guide decisions made by ptarmigan to develop the rover prototype.

### International law

The highest level of authority governing regulations is international law, specifically the outer space treaty established in 1967. Article IX states “Parties to the Treaty shall pursue studies of outer space… and conduct exploration of them so as to avoid their harmful contamination” and it states that parties must prevent “adverse changes in the environment of the Earth resulting from the introduction of extraterrestrial matter” [5]. This level of regulations is the least, however, the most important as it leads to the governing of specific rules and regulations that Ptarmigan bases their development around.

### International policies

The COSPAR Planetary Protection Policy translates the outer space treaty into planetary protection rules. COSPAR divides missions into Categories I–V based on biological risk, a rover is level IV however a sample return mission is a level V. Although the Ptarmigan rover does not return to Earth, its sample-handling functions place it within the Category V planetary protection level because according to COSPAR, “Category V missions comprise all Earth-return missions,” and returned material “shall be contained and treated as though potentially hazardous until proven otherwise” [2]. From a practical standpoint, the rover's contamination risk to mars (forward contamination) will be treated like level IV and as level V for its sample handling subsystems.

### Agency specific regulations

The most specific level of regulations are governed by the specific space agencies part of the mission, in this case NASA and ESA. In accordance with NASA and ESA planetary protection implementation frameworks, Category V requirements are flowed down to Ptarmigan. NASA policy states that “Projects shall implement planetary protection requirements appropriate to the assigned mission category” [3], therefore Ptarmigan will take strict measures to ensure level V planetary protection requirements.

Subsystems in direct contact with Mars or the sample tube and subsystems that can cause damage to sample tubes can compromise sample containment and lead to transfer of contamination into the sample that compromises its scientific integrity. Moreover, damage to the sample can cause earth to be contaminated with Martian matter, posing ecological risk.

Ptarmigan will implement a variety of solutions to uphold planetary protection policies. The sample retrieval mechanism will employ torque sensors that limit the maximum torque delivered

to prevent dents or crushing the sample tubes. The on-board sample storage system will isolate each sample. Moreover, Ptarmigan will perform cleanroom assembly, avoid porous materials, and perform Microbial assays to allow bioburden within the regulation thresholds and prevent forward contamination of mars. Moreover, Failure Modes and Effects Analysis tests such as tube drops, seal damage, and dust intrusion will be conducted to identify potential hazards and reduce them.

In accordance with IM-Mitigate principles and ESA regulations that mandate "Hazards shall be eliminated or reduced to an acceptable level" [4], risk is reduced primarily through robust engineering design rather than reliance on procedural controls alone by identifying hazards in testing and mitigating their effects.

Ptarmigan will ensure the rover maintains compliance with all levels of regulation in manufacturing and deployment while safeguarding both Martian scientific integrity and Earth's biosphere.

# Conclusion

Ptarmigan has reached the final stage of the Mars sample collection rover prototype. The sensors and the motors are wired and operational while the retrieval mechanism is still being prototyped and built. The team has selected ROS2 as the autonomy framework and chosen Nav2 for navigation and a Gazebo-based simulation environment is successfully established, running on the host VM, providing a relatively stable platform for visualizing and testing rover behavior.

# References


[1] United Nations, "Outer Space Treaty," 1967.

[2] COSPAR, "Planetary Protection Policy," 2024.

[3] NASA, "NPR 8020.12: Planetary Protection Provisions for Robotic Extraterrestrial Missions," NASA.

[4] NASA, "Spirit (Mars Exploration Rover)," NASA Science,

[5] NASA, Mars 2020, 2020, https://science.nasa.gov/mission/mars-2020-perseverance/

[6] USRA, "Tenth International Conference on Mars 2024", 2024, https://www.hou.usra.edu/meetings/tenthmars2024/pdf/3509.pdf

[7] Canadian Space Agency, "SPRP Rover", 2024, https://www.asc-csa.gc.ca/eng/multimedia/search/image/5170

# Appendix

## Preliminary Cost Estimates

The preliminary cost estimates in this report are estimated using a coalition of factors including rover size, CAD projections, MARS2020 Mission cost, as well as the Government of Canada's (GoC) CAD$1Bn budget allocation to building and maintaining a Lunar Rover.

Cost breakdown is seperated into 3 main sections:

1. Research, Development and Integration
2. Manufacturing and Assembly
3. Operational costs

Across Perseverance and Curiosity, Research, Development, and Manufacturing costs account for approximately 70-80% of a mission's lifecycle. Launch services account for approximately 10%, and operational costs a further 10%. It is important to note that launch and mission control services will be provided by UpGoers Ltd, and thus those costs have not been included in our estimates.

**Ptarmigan industries estimate a total cost of about $130,000,000.**

Our intention, as previously stated, is to build upon the CSA's SPRP prototype to reduce upfront design and testing costs. We intend to work proactively with the CSA and GoC to achieve this objective.

We also have planned for cost overruns, allocating a fund that would cover up to a 100% cost increase compared to preliminary budgets indicated here.

| Total Costing Estimate | | | |
|---|---|---|---|
| Component | Development and Integration | Manufacturing and Procurement | Lifetime Operation |
| Drivetrain and Chassis | $1,000,000 (models provided in partnership with CSA) | $50,000,000 | $0 |
| Retrieval Device and Cache | $20,000,000 | $40,000,000 | $0 |
| Software Integration | $15,000,000 | N/A | $2,000,000 |
| | | | |

Our ~$50M budget covers the improvements to the SPRP prototype as well as manufacturing the drivetrain and chassis. We also have allocated $60M to develop, implement, and manufacture a robotic arm ret0072ieval device and cache.

The ~15M software systems budget is based on a team of 20 engineers as well as data and computer scientists receiving an average yearly salary of $120K/year, during an estimated 5-year development timeline. This also includes costs for regulatory certifications.

Seeing that the rover is only intended to serve this mission, and UpGoers Ltd. Is providing the installations, mission control, and launch services to the project, we only intend to send a team of 2 rover drivers, each on an average $150K salary, and covering up to 2 years of operation on Mars. These costs are included in the “Lifetime Operation” column, as well as other unexpected costs.